STUDENT MANAGEMENT SYSTEM

Python-Based Final Project Report

Task 4: Project Documentation & Presentation — 15 Marks

Submitted By: _____

Course: BE – CSE (AIML)

College: Thakur Shyamnarayan Engineering College

Technology: Python, JSON

Academic Year: 2026–2027

1. Introduction

The Student Management System is a Python-based application designed to manage student records efficiently. It provides a simple, menu-driven interface that allows users to add, display, search, update, and delete student information.

The project uses Object-Oriented Programming (OOP) concepts to organize the code and JSON file handling to store student records permanently. It also includes input validation and exception handling to manage invalid inputs.

The main purpose of this project is to demonstrate the practical application of Python programming concepts learned during the internship.

2. Problem Statement

Maintaining student records manually can be time-consuming and may result in errors or loss of information. Searching for a particular student's details and updating records can also become difficult when information is not organized properly.

The Student Management System provides a simple digital solution to these problems. It allows users to manage student information through a menu-driven program and saves records in a JSON file for future use.

3. Project Objectives

The main objectives of this project are:

- To develop a simple student record management application.
- To provide options for adding, displaying, searching, updating, and deleting student records.
- To store student information permanently using JSON file handling.
- To validate user inputs and handle common errors.
- To implement Python fundamentals and Object-Oriented Programming.
- To develop practical programming and problem-solving skills.
- To upload and maintain the project on GitHub.

4. Features of the Project

4.1 Add Student

Allows users to enter a student's name, age, and marks.

4.2 Show All Students

Displays the details of all students stored in the system.

4.3 Search Student

Allows users to search for a student by name.

4.4 Update Student Marks

Allows users to modify the marks of an existing student.

4.5 Delete Student

Removes a student's record from the system.

4.6 File Storage

Saves student records in a file named **students.json**.

4.7 Input Validation

Checks whether the entered age and marks are valid.

4.8 Exception Handling

Handles invalid numeric inputs and common file-related errors to prevent unexpected program termination.

5. Technologies Used

Technology	Purpose
Python	Main programming language
JSON	Storing student records
VS Code	Writing and executing the program
GitHub	Hosting and submitting project files

Python Libraries and Modules

- **json:** Used to read and write student records.
- **abc:** Used to implement abstraction through an abstract base class.

Both modules are included in Python's standard library.

6. Python Concepts Implemented

The project demonstrates the following concepts:

Concept	Implementation
Variables and data types	Storing names, age, marks, and choices
Input and output	input() and print()
Type casting	int() and float()
Operators	Comparisons and arithmetic-related operations where needed

Concept	Implementation
Conditional statements	if, elif, and else
Loops	while and for loops
Functions and methods	Separate methods for each operation
Data structures	Lists and dictionaries
File handling	Reading and writing JSON files
Exception handling	try and except
Classes and objects	Student and StudentManagementSystem
Constructors	__init__()
Inheritance	Student inherits from Person
Encapsulation	Private __marks attribute
Abstraction	Abstract Person class
Polymorphism	Implementation of show_details() in the Student class

7. Project Workflow

The project follows these steps:

Step 1: The program starts and creates an object of the StudentManagementSystem class.

Step 2: The system checks whether students.json exists and loads the saved student records.

Step 3: A menu displays the available operations.

Step 4: The user selects an option.

Step 5: The program collects the required information and validates the input.

Step 6: The selected operation is performed.

Step 7: After adding, updating, or deleting a record, the program saves the changes to the JSON file.

Step 8: The menu appears again until the user selects Exit.

8. Program Output

The screenshot below shows the program running in VS Code. The saved student record is displayed using the "Show All Students" option, followed by the main menu.

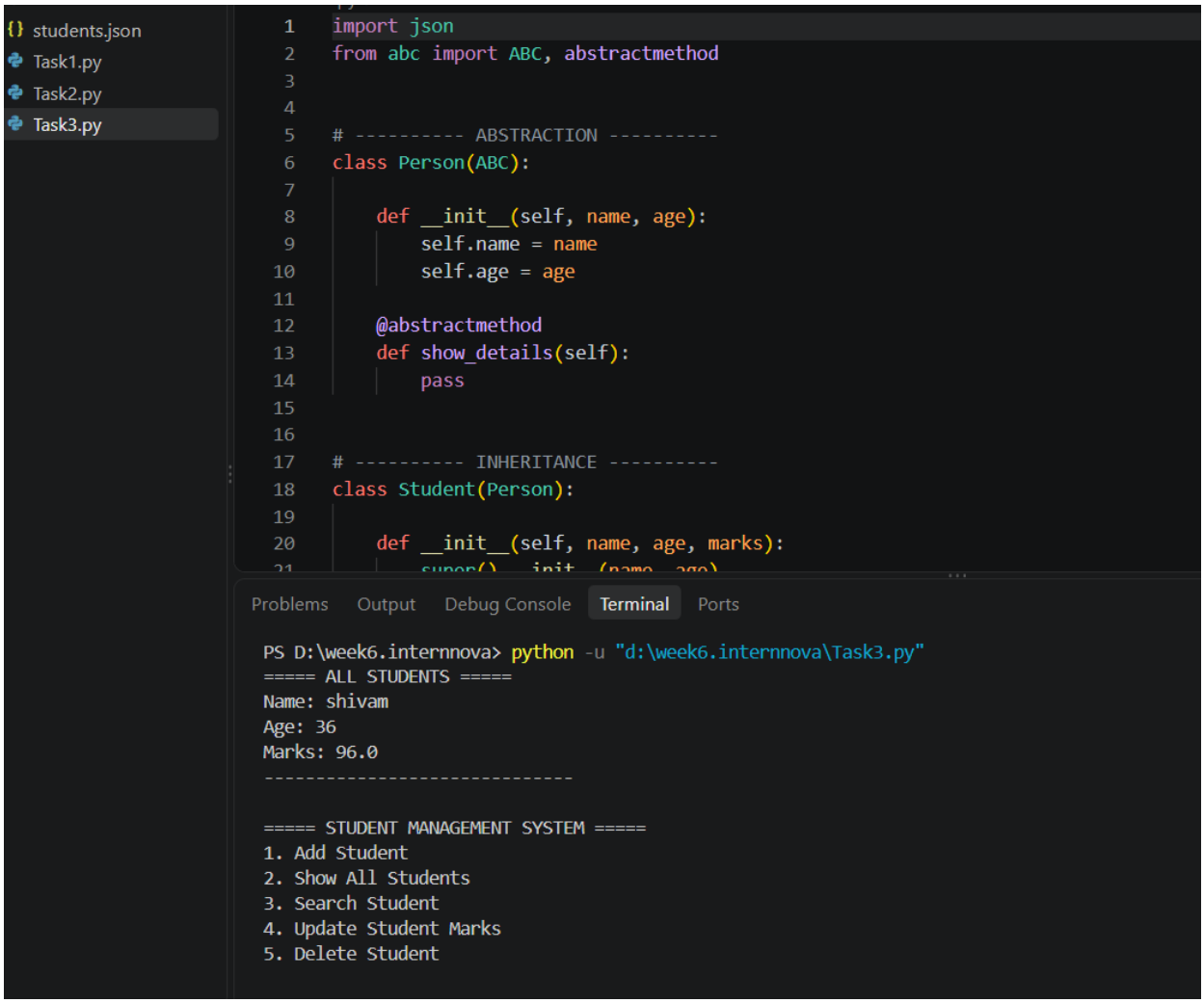

Figure 1: Source code (Task3.py) and terminal output of the Student Management System

Console output (text)

```
===== ALL STUDENTS =====
Name: shivam
Age: 36
Marks: 96.0
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===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student Marks
5. Delete Student
```